EcoTrace (MarineVigil): Multi-Modal Satellite Remote Sensing & AIS Hindcasting Engine for Marine Oil Spill Attribution

1. Executive Summary & Problem Scope

1.1 Context

Marine oil spills—whether resulting from catastrophic tanker collisions, offshore drilling blowouts, or deliberate illegal bilge-water dumping (operational discharge)—cause catastrophic and long-lasting ecological destruction. Traditionally, identifying and legally penalizing the polluting vessel in the open ocean has been nearly impossible due to:

- High ocean dynamic variability (surface currents, tides, wind drift).
- Revisit latencies in satellite remote sensing.
- Intentional evasion tactics employed by non-compliant vessels (e.g., transponder manipulation).

1.2 Objective

EcoTrace is an end-to-end, automated AI and oceanographic physics pipeline designed to:

1. Ingest multi-spectral, optical, and Synthetic Aperture Radar (SAR) imagery to segment and validate marine oil slicks while filtering false positives ("look-alikes").
2. Quantify geometric properties and simulate the slick's spatiotemporal trajectory backward (**hindcasting**) to establish a high-probability origin window, as well as forward (**forecasting**) to evaluate environmental impact.
3. Query spatial AIS vessel archives within the origin window, evaluate vessel kinematic anomalies and transponder integrity, and assign a deterministic **Vessel Suspicion Index (VSI)** score to rank culprit vessels for coast guards and maritime enforcement authorities.

2. System Architecture & High-Level Data Flow

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|                          DATA INGESTION LAYER                      |
|  [Sentinel-1 C-SAR]   [Sentinel-2/3 MSI & SST]   [ERA5 / CMEMS MetOcean]   [AIS Logs] |
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|                  AI SEGMENTATION & LOOK-ALIKE FILTERING                  |
| - Preprocessing: SAR Calibration, Speckle Filtering (Lee Filter), Terrain Masking|
| - Core Vision: U-Net / ResNet-Backbone Semantic Segmentation                  |
| - Physics-Gating Layer: SST Entropy Analysis (Algae) & Wind Vector Thresholding |
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                                      | Validated Oil Polygon
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|                HYDRODYNAMIC HINDCASTING & WEATHERING ENGINE               |
| - Physics Engine: PyGNOME (NOAA Lagrangian Particle Dispersion Model)          |
| - Inputs: Real-time/Archived CMEMS Currents (u, v) + ERA5 10m Wind Vectors      |
| - Process: Monte Carlo Reverse Advection + Evaporative Weathering Correction    |
| - Output: Spatiotemporal Origin Probability Distribution (Bounding Box & Time)  |
+-----------------------------------------+-------------------------------------------+
                                      | Origin Polygon P(x, y, t)
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|                  VESSEL ATTRIBUTION & SCORING ENGINE                  |
| - Database: PostGIS Spatial Queries (ST_Intersects, ST_DWithin)               |
| - Trajectory Reconstruction: Linear/Cubic Spline Interpolation for AIS Gaps      |
| - Multi-Factor Scoring: 4-Factor Weighted Composite Suspicion Index (VSI)       |
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                                      | Ranked Suspects List & Evidence
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|                  ENFORCEMENT & VISUALIZATION DASHBOARD                  |
| - Interactive Mapbox GL UI, Time-Slider Replay, Trajectory Overlay, PDF Dossiers |
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3. Comprehensive Solutions to Edge Cases and Vulnerabilities

Edge Case 1: Natural Hydrocarbon Seeps

- **Problem:** Natural geological seeps emit crude oil continuously from the seabed, mimicking man-made spills on SAR but having no human culprit.
- **Solution (Spatial Masking & Temporal Anomaly Gating):**
  - Integrate global seep databases (e.g., NOAA/USGS benthic hydrocarbon seep catalog)

as a static geo-spatial overlay layer in PostGIS.
    - Compute spatial intersection: if a spill candidate falls within radius R_{seep} ($<2\text{ km}$) of a documented seep coordinate, it is tagged as NATURAL_SEEP_CANDIDATE.
    - The system computes a temporal recurrence metric: natural seeps appear repeatedly in the same coordinates across multiple satellite acquisitions over months, whereas vessel discharges are one-off spatial anomalies.

Edge Case 2: Heavy Rain Cells & Low-Wind Slicks

- **Problem:** Rain cells cause atmospheric attenuation and ocean surface smoothing ("rain dampening"), creating dark SAR patches. Wind speeds below 3 m/s leave the sea smooth (no capillary waves), generating widespread false positives.
- **Solution (Met-Ocean Gating Filter):**
    - Ingest ECMWF ERA5 hourly total precipitation and surface wind speed data.
    - Establish hard rejection thresholds:
      $$\text{Status} = \begin{cases} \text{REJECT (Low Wind)}, & \text{if } \Vert{}\vec{V}_{\text{wind}}\Vert{} < 3.0\text{ m/s} \\ \text{REJECT (High Wind Dissipation)}, & \text{if } \Vert{}\vec{V}_{\text{wind}}\Vert{} > 14.0\text{ m/s} \\ \text{REJECT (Rain Artifact)}, & \text{if } P_{\text{rain}} > 5.0\text{ mm/hr} \\ \text{ACCEPT TO ML PIPELINE}, & \text{otherwise} \end{cases}$$

Edge Case 3: Sun Glint on Optical Sensors

- **Problem:** Direct specular reflection of solar radiation blinds optical sensors (Sentinel-2 MSI), washing out spectral features.
- **Solution (Sensor Complementarity & Geometric Masking):**
    - Use Sentinel-1 SAR as the primary detection medium (active microwave radar is immune to sun glint, clouds, and night conditions).
    - For optical validation pipelines, compute the Sun Glint Angle θ_g using Solar Zenith Angle (θ_s), Satellite Zenith Angle (θ_v), and Relative Azimuth Angle (ϕ):
      $$\cos \theta_g = \cos \theta_s \cos \theta_v - \sin \theta_s \sin \theta_v \cos \phi$$
    - Pixels where $\theta_g < 30^\circ$ are automatically masked out of the optical processing pipeline to eliminate saturation artifacts.

Edge Case 4: Multiple Overlapping & Merging Spills

- **Problem:** Discharges from distinct vessels can merge under current aggregation, creating a single contiguous slick that skews the physical center of mass.
- **Solution (Morphological Decomposition & Plume-Tail Tracing):**
    - Instead of calculating a single global centroid, apply distance-transform watershed segmentation and directional skeletonization to the binary mask.
    - Extract the linear "tails" (thinner, more weathered extremities) versus the thicker "heads" (recent discharge centers).
    - PyGNOME executes independent multi-seed hindcast simulations starting from each

isolated skeleton terminus, generating distinct origin trajectories rather than one false averaged coordinate.

Edge Case 5: Sub-surface Suspended / Weathered Oil

- **Problem:** Heavy or weathered fuel oil can sink just below the surface wave crests, disappearing from SAR backscatter detection while continuing to travel with currents.
- **Solution (Multi-Spectral Water-Column Penetration):**
  - Ingest Sentinel-2 Level-2A multi-spectral bands (Band 2 Blue: 490 nm, Band 3 Green: 560 nm, Band 4 Red: 665 nm, Band 8 NIR: 842 nm).
  - Calculate the **Fluorescence Index (FI)** and **Floating Algae Index (FAI)** adapted for hydrocarbons:
    $$\text{FAI} = R_{\text{rc, NIR}} - \left(R_{\text{rc, RED}} + (R_{\text{rc, SWIR}} - R_{\text{rc, RED}}) \times \frac{\lambda_{\text{NIR}} - \lambda_{\text{RED}}}{\lambda_{\text{SWIR}} - \lambda_{\text{RED}}} \right)$$
  - The differential absorption and subsurface upwelling radiance profile identify suspended hydrocarbon plumes down to 2–5 meters in clear-to-moderate ocean water.

Edge Case 6: Extreme Weathering and Evaporative Mass Loss

- **Problem:** Volatile fractions (e.g., marine diesel) evaporate within hours, altering the slick geometry and making the detected area misleadingly small relative to the original spill volume.
- **Solution (Integrated ADIOS Weathering Physics Engine):**
  - Couple NOAA's Automated Data Inquiry for Oil Spills (ADIOS) library inside PyGNOME.
  - Run backward mass-balance integration accounting for sea temperature, solar irradiance, and wind-driven emulsification:
    $$M(t) = M_0 \cdot \exp(-k_{\text{evap}} \cdot \sqrt{t})$$
  - The system reconstructs the initial discharge volume and footprint boundary, preventing underestimation of the spill origin polygon.

Edge Case 7: Complex Coastal Hydrodynamics & Shallow-Water Tides

- **Problem:** Open-ocean global circulation models (e.g., $1/12^\circ$ standard ocean current grids) fail inside coastal bays, ports, and estuaries due to shallow bathymetry and reciprocating tidal currents.
- **Solution (Hierarchical Hydrodynamic Model Switching):**
  - Implement an automatic distance-to-shore trigger:
    - **Offshore ($>20\text{ km}$ from baseline):** Ingest global CMEMS/HYCOM surface currents ($1/12^\circ$ resolution).
    - **Coastal / Inshore ($\le 20\text{ km}$ from baseline):** Dynamically switch the particle tracking engine to high-resolution local models (e.g., ADCIRC 2D/3D tidal hydrodynamic mesh or high-frequency coastal radar grids).

Edge Case 8: Satellite Revisit Latency & Temporal Gaps

- **Problem:** Long intervals between satellite passes mean a slick could be detected 24 to

72 hours after dumping, causing the backward-projected origin cone to expand over a massive geographic area.

- **Solution (Monte Carlo Probabilistic Cones & Dynamic Score Normalization):**
  - Avoid single-point hindcasting. Inject turbulent diffusion noise into the advection equation using a random walk model:
    $$dX_t = \vec{u}(\vec{x}, t) dt + \sqrt{2 K_h dt} \, \xi$$
    (where K_h is the horizontal eddy diffusivity and $\xi \sim \mathcal{N}(0, 1)$).
  - Simulate $N = 10,000$ particles in reverse to output a calibrated 2D Probability Density Function (PDF) map of the origin window.
  - Adjust candidate certainty thresholds: if the 95% confidence origin polygon spans $>500\text{ km}^2$, candidate VSI scores are mathematically scaled to indicate lower attribution certainty.

4. Vessel Attribution Engine & Anomaly Detection

4.1 Resolving "Dark Vessels" (AIS Transponder Disabling)

When non-compliant vessels disable their Class-A AIS transponders to dump oil undetected:

1. **Dead-Reckoning & Kinematic Interpolation:** When a transponder ping gap exceeds 30 minutes, the engine computes a spline trajectory connecting the last broadcast point (P_{last}, t_1) to the re-acquisition point (P_{next}, t_2).
2. **Speed & Feasibility Gate:** The system evaluates whether the direct route across the gap intersects the hindcast origin polygon within the expected time window $[t_{\text{spill}} - \Delta t, t_{\text{spill}} + \Delta t]$ based on the vessel's cruising speed profile.
3. **Ghost Penalty:** Any vessel with an unexplained AIS drop within 50 km of the spill origin during the calculated time window is flagged with maximum evasion weight.

4.2 Congested Shipping Lanes (Separating Culprits from Innocent Traffic)

In high-density corridors (e.g., Strait of Malacca, English Channel), dozens of vessels intersect the origin polygon. The system separates the offender using kinematic and behavioral metrics:

- **Speed Drop Profile:** Operational discharge of heavy engine sludge/bilge water is hydrodynamically ineffective at high cruising speeds and is typically conducted at reduced speeds ($2\text{ to }6\text{ knots}$).
- **Course Deviations:** Sudden zigzagging or circular holding patterns away from standardized Traffic Separation Schemes (TSS).

5. Mathematical Formulation: Vessel Suspicion Index (VSI)

The overall suspicion score for any candidate vessel i is defined by a normalized composite index:

$$VSI_i = w_1 F_{\text{loc}}(i) + w_2 F_{\text{spd}}(i) + w_3 F_{\text{gap}}(i) + w_4 F_{\text{prof}}(i)$$

$$\text{Subject to: } \sum_{j=1}^4 w_j = 1.0 \quad (w_1=0.40, \; w_2=0.25, \; w_3=0.25, \; w_4=0.10)$$

VESSEL SUSPICION INDEX (VSI) BREAKDOWN		
Metric Component	Weight	Mathematical Formulation
Spatiotemporal Proximity (F_loc)	40%	F_loc = exp(- d_min^2 / (2 * sigma_model^2))
Kinematic Anomaly (F_spd)	25%	F_spd = max(0, (v_nominal - v_min)/v_nominal)
AIS Evasion Gap (F_gap)	25%	F_gap = min(1.0, Delta_t_gap / 120 minutes)
Vessel Profile Risk (F_prof)	10%	F_prof = Discrete lookup (Tanker=1.0, Bulk=0.7)

Component Formulations

1. Spatiotemporal Proximity (F_{loc})

Computed via Gaussian decay over the Euclidean minimum distance d_{min} between the vessel track and the Monte Carlo hindcast origin centroid (μ_x, μ_y) at estimated discharge time t_0:

$$F_{\text{loc}} = \exp\left(-\frac{d_{\text{min}}^2}{2\sigma_{\text{model}}^2}\right)$$

(where σ_{model} is the standard deviation radius of the particle cluster at t_0).

2. Kinematic / Deceleration Anomaly (F_{spd})

Measures anomalous deceleration relative to the vessel's nominal operating speed:

$$F_{\text{spd}} = \max\left(0, \; \frac{v_{\text{nominal}} -

v_{\text{origin}}}{v_{\text{nominal}}}\right)$$

3. AIS Transponder Evasion Factor (F_{gap})

Penalizes transponder outages occurring within the bounding zone:

$$F_{\text{gap}} = \min\left(1.0, \; \frac{\Delta t_{\text{gap}}}{120\text{ min}}\right)$$

4. Vessel Risk Profile Matrix (F_{prof})

Mapped deterministically from AIS static message metadata (ITU-R M.1371):

$$F_{\text{prof}} = \begin{cases} 1.0, & \text{Crude Oil / Chemical / Product Tankers} \\ 0.7, & \text{Large Container Ships / Bulk Carriers (Heavy Fuel Oil bilge risk)} \\ 0.3, & \text{Tugs / Offshore Supply Vessels} \\ 0.1, & \text{Fishing Vessels} \\ 0.0, & \text{Passenger Ferries / Sailing Yachts} \end{cases}$$

6. End-to-End Technology Stack & Data Pipelines

Sub-system	Technology / Library	Purpose
Satellite Imagery	Sentinel-1 SAR (GRD, C-Band), Sentinel-2 MSI, Sentinel-3 SLSTR	Multi-sensor dark-patch segmentation, optical confirmation, Sea Surface Temperature.
Met-Ocean Feeds	Copernicus CMEMS API, ECMWF ERA5	Surface current velocity vectors (u, v), 10m wind fields (u_{10}, v_{10}), precipitation, wave height.
Vessel Data	MarineCadastre / AISHub / Spire API	Historical and live dynamic (lat, lon, SOG, COG, timestamp) and static (MMSI, IMO, ShipType) logs.
Computer Vision	PyTorch, torchvision, OpenCV, GDAL	Deep learning semantic segmentation (U-Net with EfficientNet-B4 backbone), SAR despeckling.

Physics Simulation	**PyGNOME** (NOAA), ADIOS Oil Library	Backward/forward Lagrangian particle transport, oil weathering, evaporation, and dispersion.
Spatial Database	**PostgreSQL 16 + PostGIS 3.4**	Spatial indexing (R-Tree on geometry(LineString)), spatio-temporal queries (ST_DWithin, ST_Intersects).
Backend API	Python 3.11, FastAPI, Celery, Redis	Asynchronous pipeline orchestration, heavy simulation job queue, high-performance API endpoints.
Frontend / UI	React.js, TypeScript, **Mapbox GL JS** / Deck.gl	Multi-layer GPU map rendering, particle animation, interactive timeline playback, suspect dossier generation.

7. Implementation Blueprint: Execution Plan for SIH

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| PHASE 1: DATA PREPARATION & ML (Hours 0 - 12)                                      |
| - Download Zenodo Sentinel-1 SAR Oil Spill dataset & sample MarineCadastre AIS logs. |
| - Train / fine-tune lightweight U-Net model on SAR patches; export ONNX runtime
model.|
| - Build PostGIS schema and ingest AIS historical trajectories with spatial indices. |
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| PHASE 2: PHYSICS & HINDCASTING INTEGRATION (Hours 12 - 24)                    |
| - Integrate PyGNOME scripts to read netCDF ocean current/wind fields.              |
| - Implement reverse particle advection function with Monte Carlo distribution output.  |
| - Connect AI mask output bounding box -> PyGNOME initialization seed points.         |
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| PHASE 3: SCORING ENGINE & ATTRIBUTION (Hours 24 - 32)                          |
| - Write spatial SQL queries joining vessel tracks with the calculated origin polygon. |
| - Implement the 4-Factor VSI formula in FastAPI backend.                        |
| - Generate ranked output list with individual sub-score breakdowns.               |
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| PHASE 4: UI/UX DASHBOARD & POLISH (Hours 32 - 36)                              |
| - Build Mapbox GL frontend displaying satellite mask overlays & ship track playback.  |
| - Add suspect evidence panel showing MMSI, IMO, VSI Score, and anomalous speed
graph. |
| - Rehearse live demonstration using a known historical spill case.                |
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8. Feasibility, Viability, and Strategic Risk Management

Feasibility Analysis

- **Data Accessibility:** Copernicus Sentinel SAR data and CMEMS oceanographic feeds are freely accessible under the EU open-access charter. MarineCadastre provides complete open AIS datasets.
- **Compute Footprint:** Pre-trained U-Net inference takes $<2\text{ seconds}$ on a standard GPU. PyGNOME Lagrangian trajectory simulation with $10,000$ particles runs in $<5\text{ seconds}$ on modern multi-core CPUs.

Viability & Target Stakeholders

- **Coast Guard & Maritime Law Enforcement:** Automates evidence gathering for maritime violations under MARPOL 73/78 Annex I regulations.
- **Port Authorities & Maritime Insurance:** Provides deterministic liability proof to recover multi-million-dollar coastal cleanup expenses from offending shipping carriers.

Risk Management Matrix

RISK MANAGEMENT MATRIX		
Identified Risk	Severity	Technical Mitigation Strategy
High Cloud Cover / Bad Optical Weather	High	C-Band SAR (Sentinel-1) penetrates all cloud cover and functions in complete darkness.
Complex Chaotic Ocean Turbulence	Medium	Monte Carlo dispersion simulation outputs calibrated probabilistic envelopes rather than single points.
Disconnected / Dark AIS Transponders	High	Spline trajectory interpolation across transponder gaps coupled with severe suspicion score penalties.
Spatial Query Database Bottlenecks	Medium	Spatial partitioning (`BRIN` / `GIST` indices) in PostGIS bounding searches by time and geographic tile.

9. Academic & Research References (For Pitch Presentation)

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2. **Lagrangian Drift Modeling & Hindcasting:**
   - Zodiatis, G., Lardner, R., et al. (2017). *Advanced Oil Spill Modeling with MEDSLIK*. **Journal of Marine Research**, 75(6), 841–869. *(Supports the integration of reverse advection with met-ocean Eulerian velocity grids)*.
3. **AIS Anomaly Detection & Vessel Tracking:**
   - de Vries, G. K. D., & van Someren, M. (2012). *Machine learning for vessel trajectories using compression, alignments and clustering*. **IEEE Transactions on Intelligent Transportation Systems**, 13(3), 1192–1204.
4. **Physical Modeling Framework:**
   - Beegle-Krause, C. J. (2001). *General NOAA Operational Modeling Environment*

(GNOME): A new tool for oil spill modeling. **International Oil Spill Conference Proceedings**, 2001(2), 1477–1481.